

25CSCN08I Local/Wide Area Networks and Network Protocols

Design and Simulation of a Local Area Network (LAN) for a University Building Using Cisco Packet Tracer

Phase 2 Final Report

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Abstract

I This report is about designing and simulating a LAN for a university building in Cisco Packet Tracer. The network follows a three-layer hierarchical model with a core Layer 3 switch for inter-VLAN routing, two distribution switches, and four access switches. I made four VLANs to keep departements seperated — IT, Admin, Student WiFi, and Server Room. Theres a DHCP server that gives IPs automaticly, plus DNS, HTTP and FTP services. Student laptops connect wirelessly through WPA2-PSK. All tests came back succesful.

Introduction

Nowdays universities depend alot on there networks. Students need WiFi for lectures, staff need connections for shared drives, and IT needs its own seperate segment so test machines dont mess with production traffic. If the LAN isnt designed properly it causes congestion and security problems. Forouzan [1] says a well-structured LAN is the backbone of any organization and after this project i can see why.

In Phase 1 i submitted a Project Proposal omar239372 with the goals and genral approach. This Phase 2 report is basicly the follow-through — I built the network, configured evrything, and tested it. The sections below cover my topology, addressing, commands, and test results.

Objectives

Based on the coursework brief for module 25CSCN08I, these are the objectives i was trying to acheive:

- Design a LAN topology that can work for a university building with multiple departements.
- Use VLANs to logically seperate traffic between IT, Admin, students and the server room.
- Configure DHCP so that IP addressing is automatic and i dont have to type IPs manually.
- And I set up DNS and web services that are acessable across all vlans
- Now I did Provided secure wireles for student laptps using an access point with WPA2 PSK.
- So I Test and document evrything with screnshots to prove it all works proper and that's what I did.

Scope of Work & Methodology

The whole project was simulated in Cisco Packet Tracer 8.2, no physical equipment at all. I sketched the topology on paper first then placed devices in Packet Tracer and cabled them up. After that i configured from the core outward, set up servers, and finally tested evrything.

Networking Media & Cabling Best Practices

For cabling buildings theres basicly two options. Copper (Cat5e/Cat6 UTP) is standard for horizontal runs becuase its cheep and supports Gigabit up to 100 metres per IEEE 802.3 [3]. The problem is it picks up interference near power cables so cable managment matters.

Fibre optic is better for backbone links between floors becuase it handles longer distances and is immune to EMI, but costs more [2]. In my simulation i used copper FastEthernet since evrything is on one floor, but in a real deployment i would use fibre for core-to-distribution uplinks as Tanenbaum [2] reccomends.

Network Topology Design

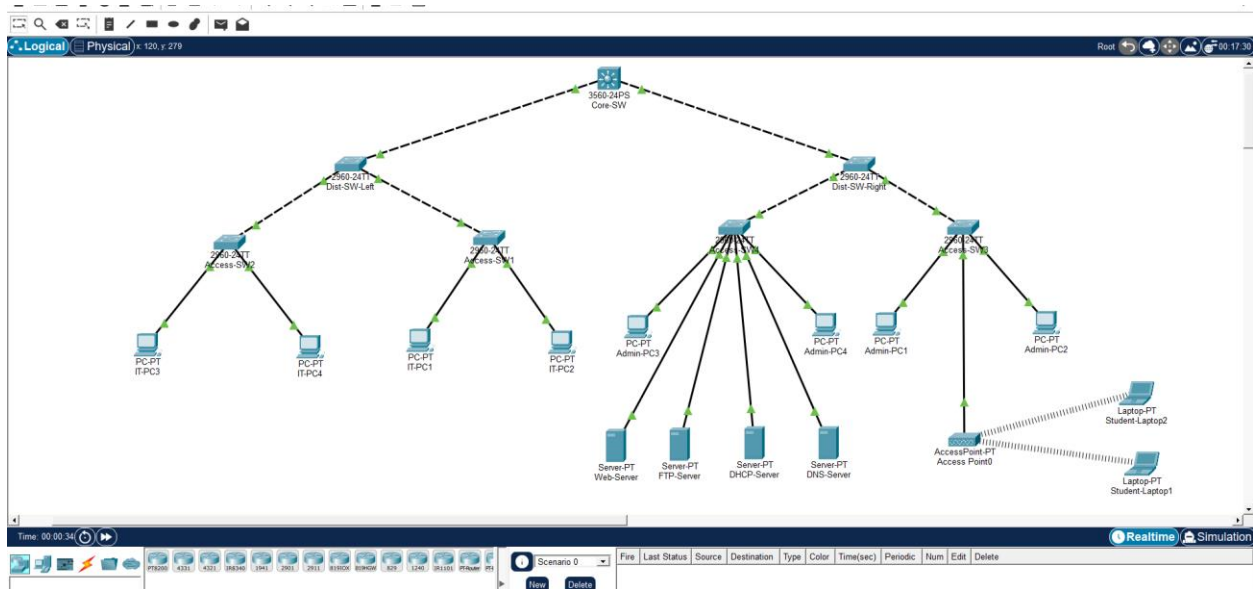


Figure 0 — Full network topology in Cisco Packet Tracer showing the three-layer hierarchical design.

I went with the Cisco three-layer hierarchical model because that's what we covered in lectures. In a flat network broadcasts flood everywhere and it's hard to find problems. Three layers keep things organized.

For the Core Layer One 3560-24PS multilayer switch I added CoreSw with IP routing enabled, and it forwards packets between VLANs using SVIs as I said and did Distribution Layer Two 2960-24TT switches (Dist-SWLeft and Dist-SW-Right) that aggregate uplinks over 802.1Q trunks. Access Layer Four switches (Access-SW1 to SW4) connecting PCs, servers, and the AP, each port in the right VLAN.

I considered a flat topology but Kurose and Ross [4] say hierarchical is better for anything bigger than a small office and I agree, the layered approach made troubleshooting way easier.

IP Addressing Scheme

For the addressing I used the 192.168.x.0/24 private range and split it across four VLANs. Each /24 subnet gives 254 useable host addresses which is more than enough for this simulation and also leaves room if more devices need to be added later on.

VLAN ID	Name	Network	Gateway	Subnet Mask
10	IT_Dept	192.168.10.0/24	192.168.10.1	255.255.255.0
20	Admin_Dept	192.168.20.0/24	192.168.20.1	255.255.255.0
30	Student_WiFi	192.168.30.0/24	192.168.30.1	255.255.255.0
40	Server_Room	192.168.40.0/24	192.168.40.1	255.255.255.0

The server addresses in VLAN 40 are shown below:

Device	IP Address
DHCP-Server	192.168.40.10
DNS-Server	192.168.40.20
Web-Server	192.168.40.30
FTP-Server	192.168.40.40

Device Configuration Details

This section shows the main CLI commands i typed into the devices. Im not going to list every single line from the running config, just the important parts.

Core Switch (Core-SW)

First thing i did was give it a hostname and set an enable secret password. I also added a banner message and turned on password encryption:

```
enable
configure terminal
hostname Core-SW
enable secret cisco123
banner motd #WARNING: Authorized Access Only!#
service password-encryption
```

Then i created the four VLANs and set up the trunk ports on Fa0/1 and Fa0/2 using 802.1Q [3]:

```
vlan 10
 name IT_Dept
vlan 20
 name Admin_Dept
vlan 30
 name Student_WiFi
vlan 40
 name Server_Room
!
interface range fa0/1 - 2
 switchport trunk encapsulation dot1q
 switchport mode trunk
```

The most important part was enabling Layer 3 routing and creating the SVIs so the VLANs can actually talk to each other:

```
ip routing
!
interface vlan 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
interface vlan 20
 ip address 192.168.20.1 255.255.255.0
 no shutdown
interface vlan 30
 ip address 192.168.30.1 255.255.255.0
 no shutdown
interface vlan 40
 ip address 192.168.40.1 255.255.255.0
 no shutdown
```

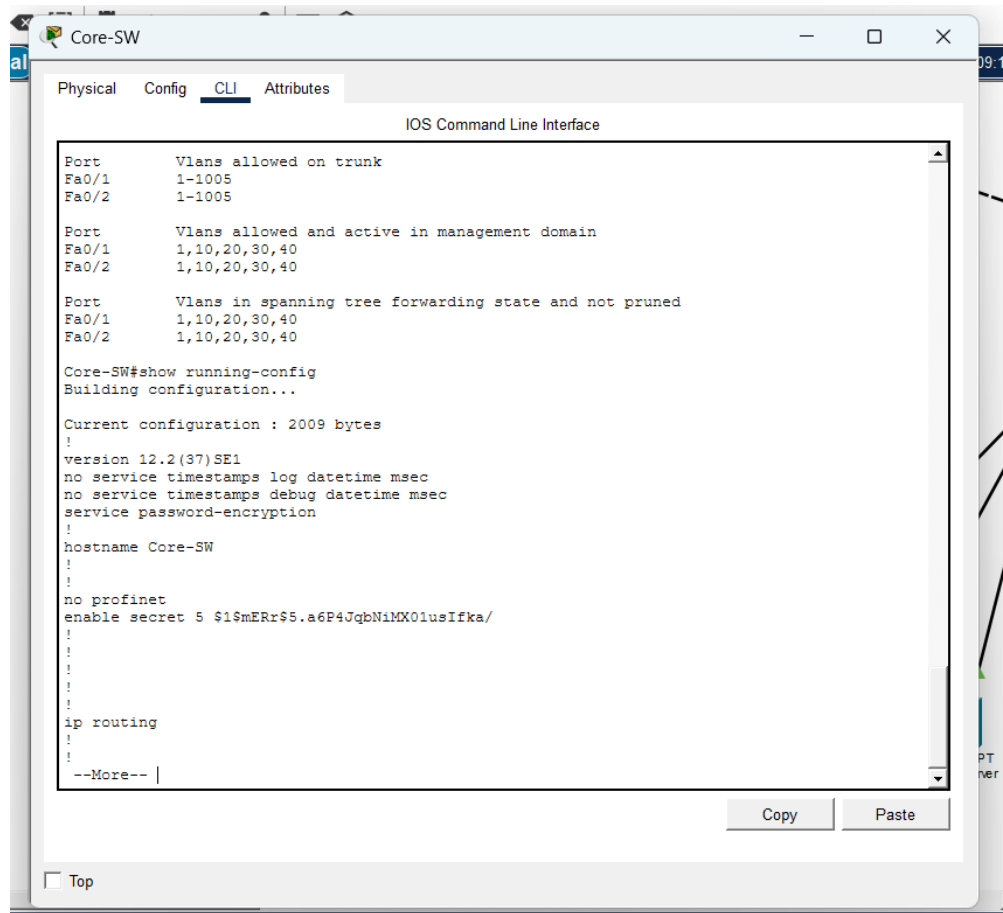


Figure 1 — Core-SW running config showing hostname, enable secret and ip routing enabled.

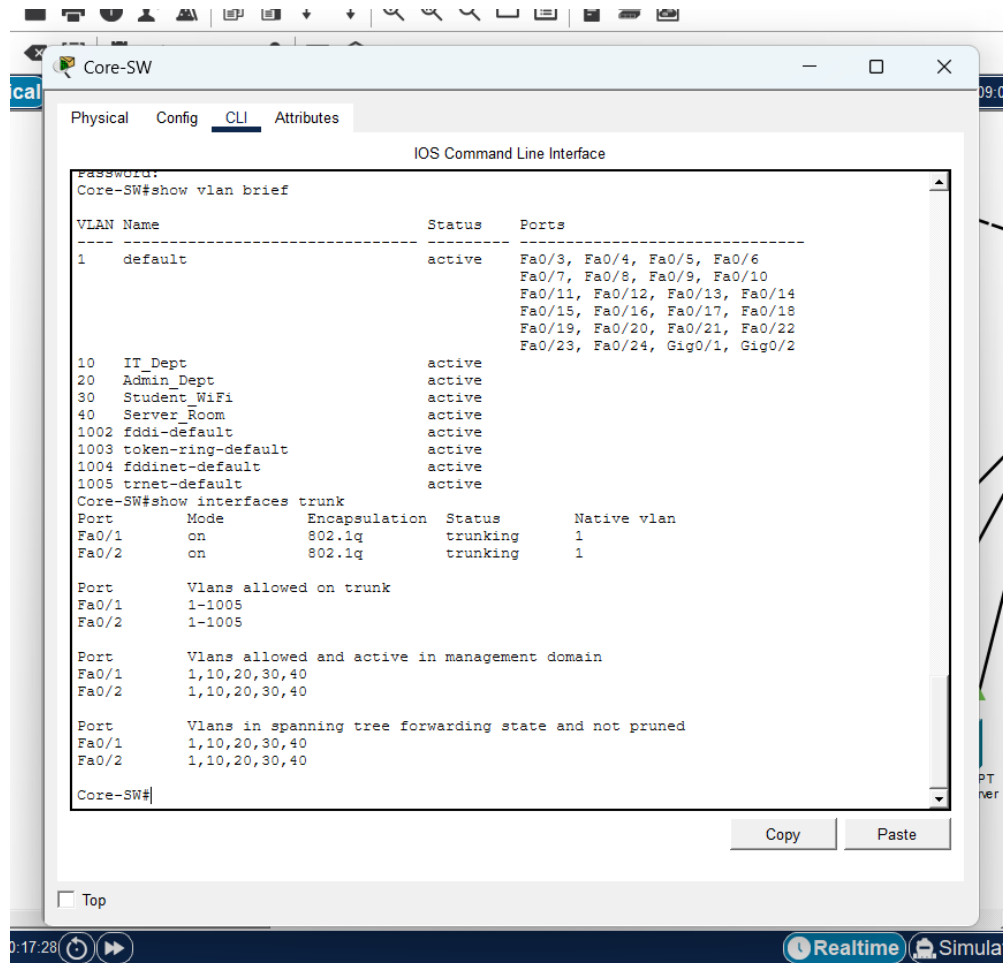


Figure 2 — Core-SW show vlan brief and show interfaces trunk. Fa0/1 and Fa0/2 are trunking via 802.1Q.

Distribution & Access Switches

The other switches were configured similarly — each got a hostname, enable secret, same VLANs, and trunks going up. On Access-SW1 for example Fa0/1 and Fa0/2 are in VLAN 10 and Fa0/3 is the trunk uplink.

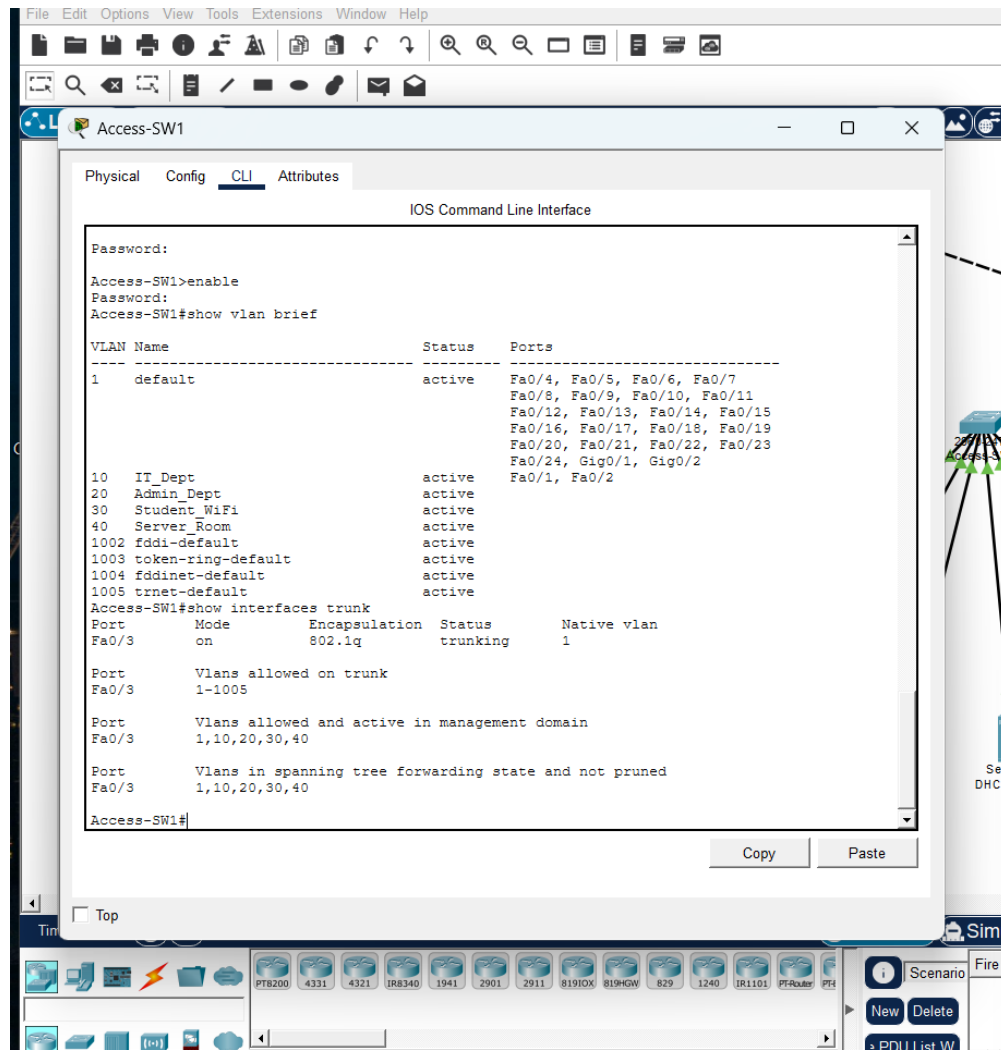


Figure 3 I did th Acces SW1 showing VLANs and trunk configuration. Fa0/1-2 in VLAN 10 Fa0/3 trunkng

DHCP, DNS & Web Servers

My DHCP server 192.168.40.10 has pools for VLAN 10, 20 and 30 with the correct gateways and DNS. DNS server (192.168.40.20) maps www.university.local to 192.168.40.30 which is the Web server FTP server at 192.168.40.40 uses credentials cisco/cisco

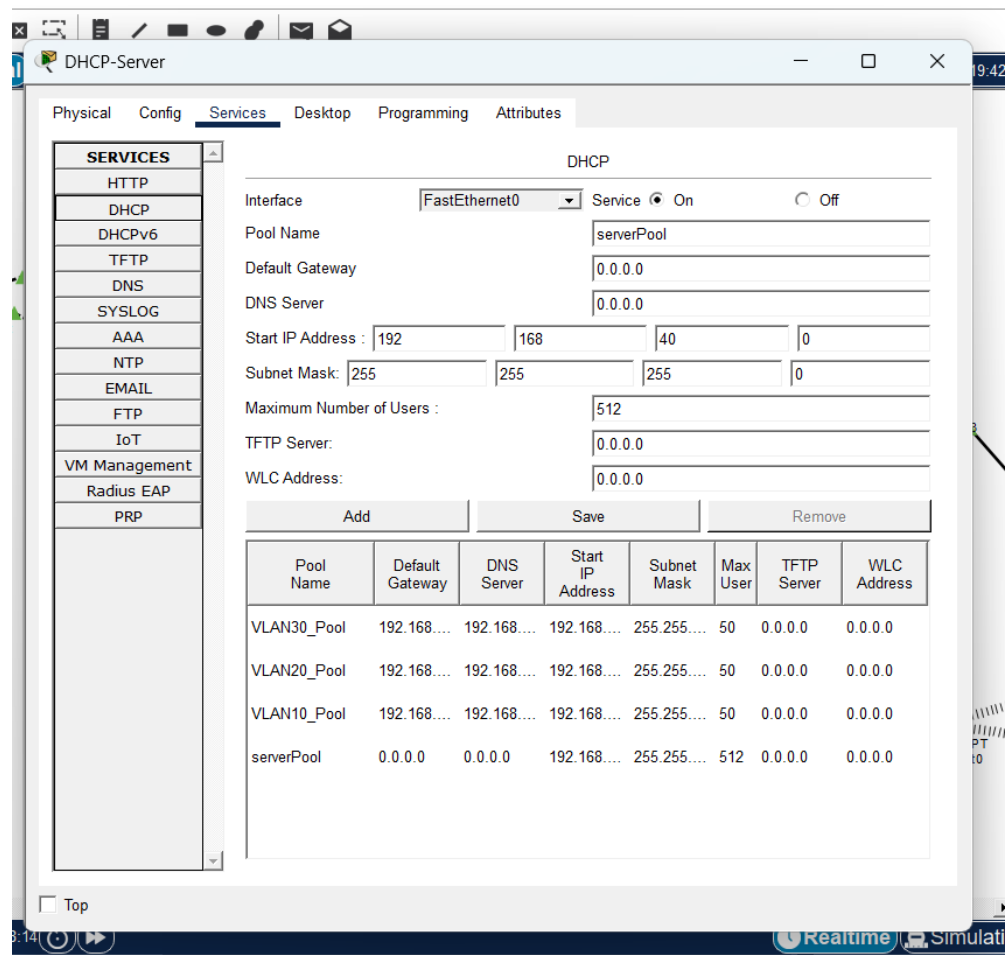


Figure 4 my DHCP-Server showing the pools for VLAN10, VLAN20, and VLAN30.

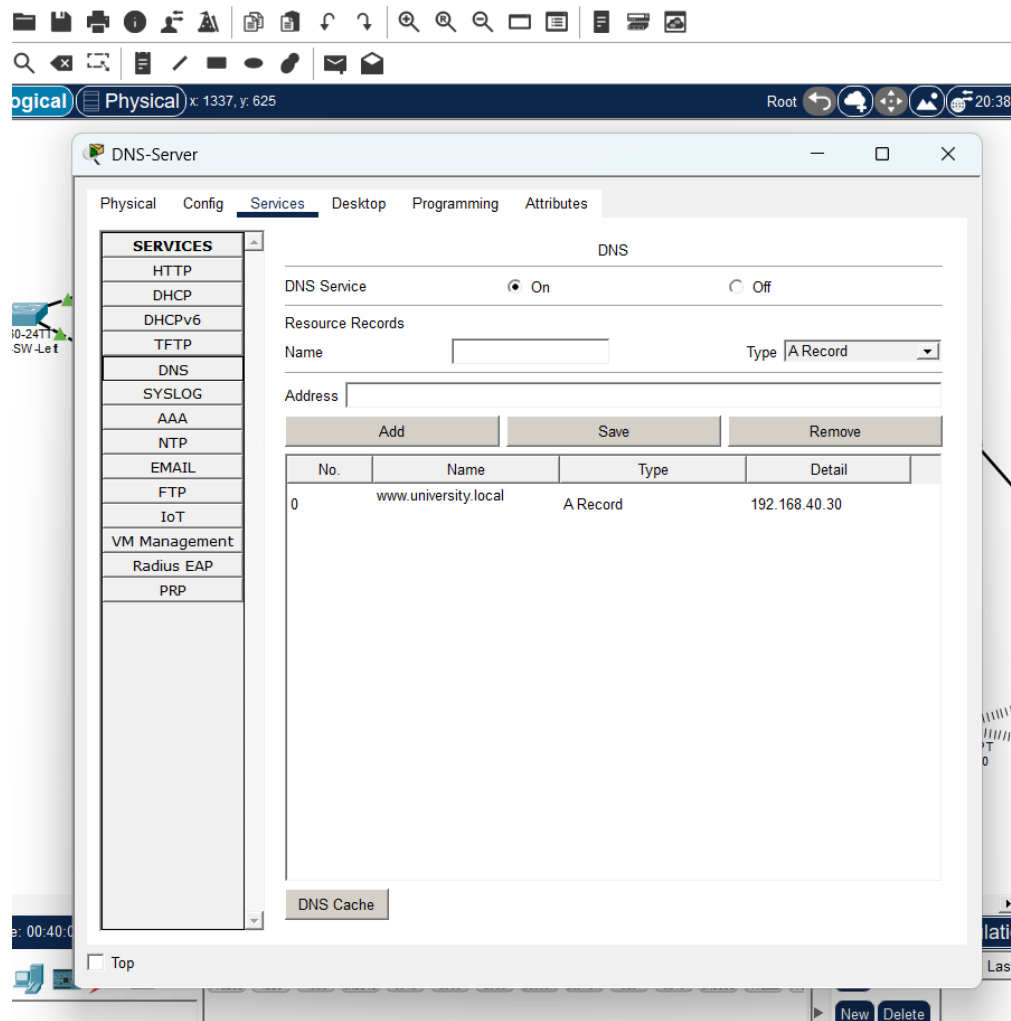


Figure 5 now myDNS Server itsb with A record for `www.university.local` pointing to `192.168.40.30`.

Wireless Access Point

And I think For the student laptops i set the access point SSID to "University-WiFi" on channel 6 Security is WPA2-PSK with AES encryption and the pasphrase is "university2025" The AP connects to a switch prt thats in VLAN 30 [Student_WiFi] so laptops that join the WiFi automaticly end up in the right VLAN This follws the IEEE 802.11 wireless standards [5].

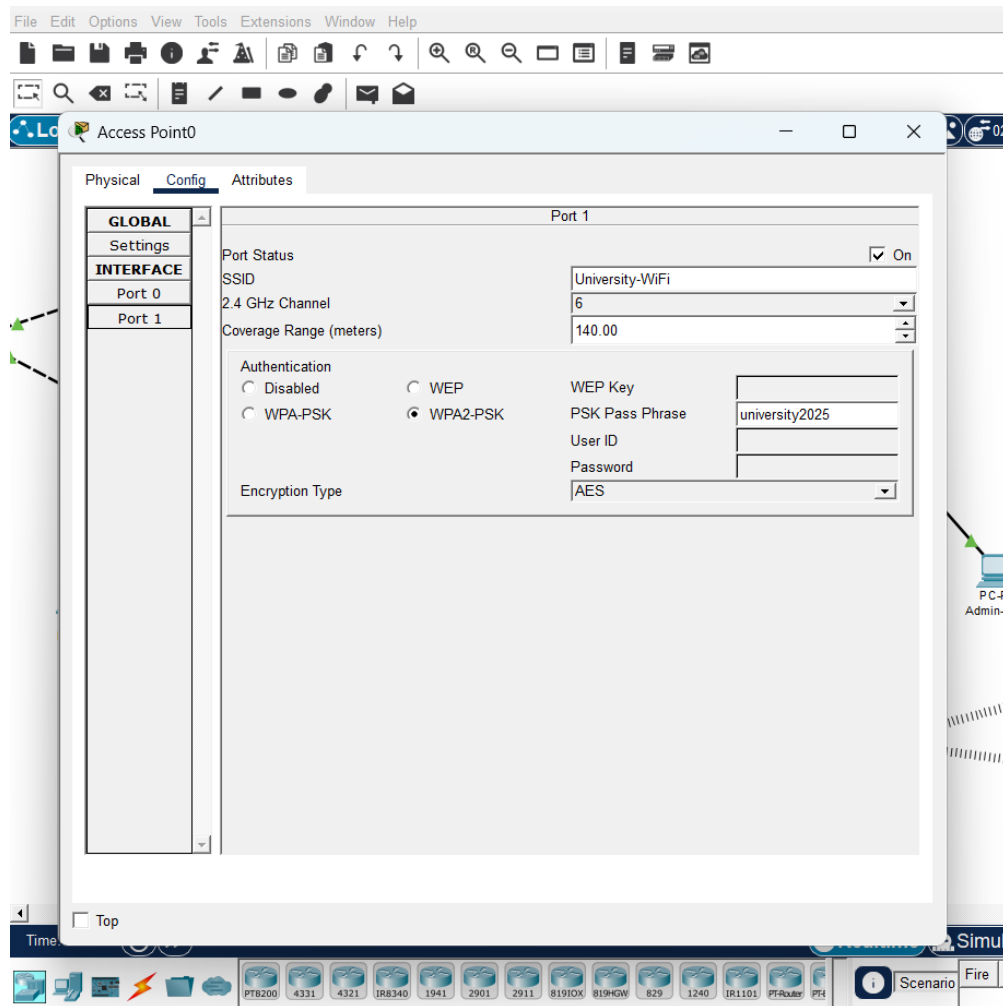


Figure 6 soothss Access Point config: SSID University-WiFi andd WPA2-PSK AES encryptinn, channell 6.

Result & Testing

After configuring evrything i ran a bunch of tests to make sure the network actualy works.

DHCP Verification

I opend IP Configuration on multiple PCs. IT PC3 got 192.168.10.101 with gateway 192.168.10.1 and DNS 192.168.40.20 which is corect for VLAN 10. Admin-PC3 got 192.168.20.104 and showed "DHCP request successful" so the server is working across VLANs.

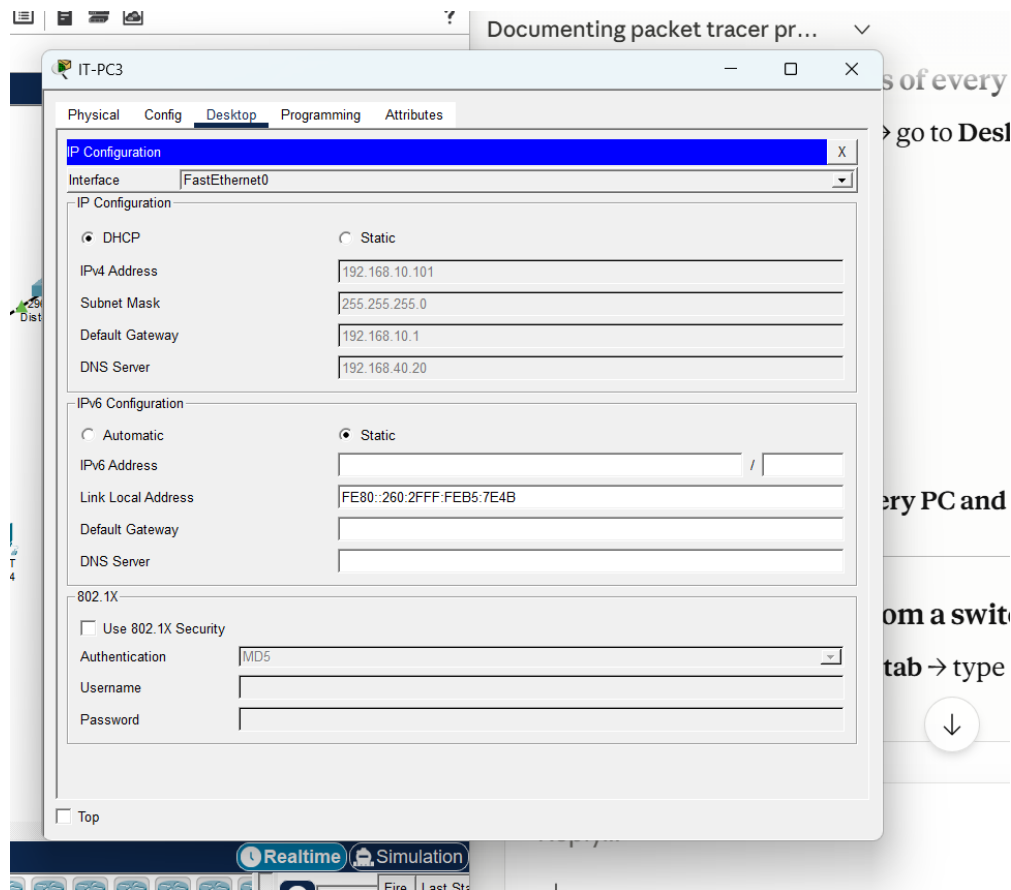


Figure 7 — IT-PC3 showing DHCP assigned IP 192.168.10.101 with correct gateway and DNS.

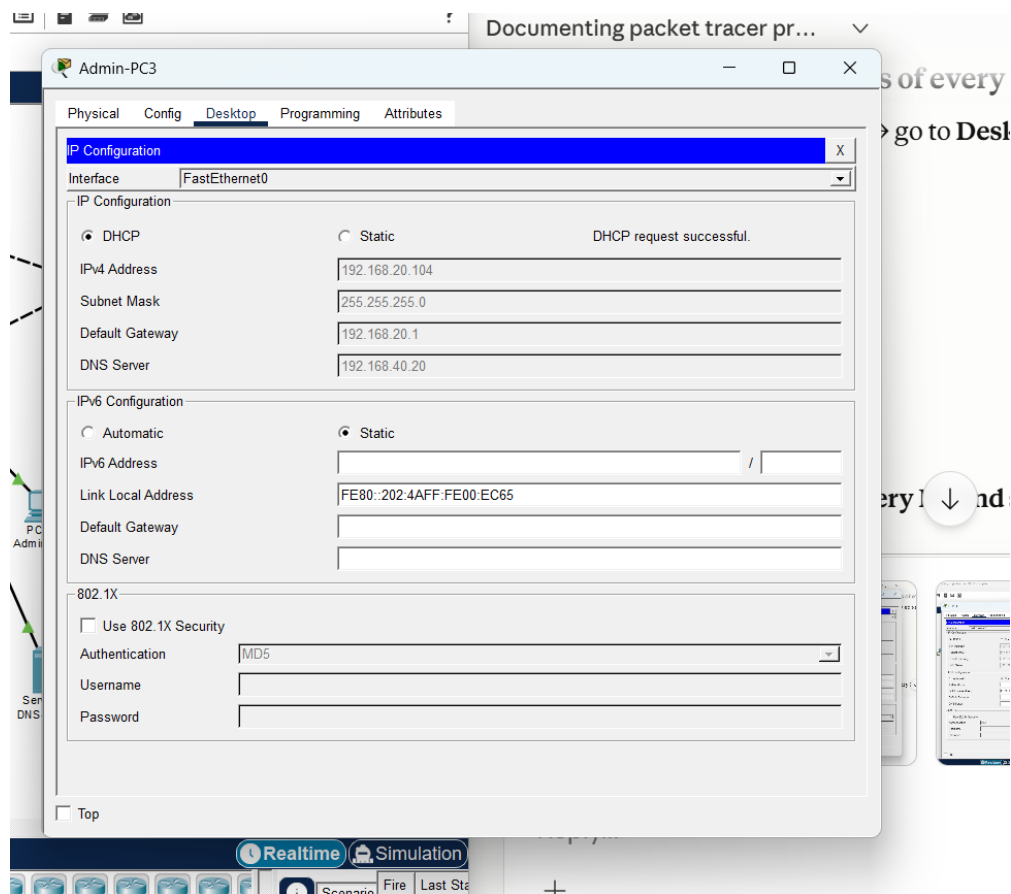


Figure 8 Admin-PC3 showing DHCP assigned 192.168.20.104 and DHCP request successful message.

Inter-VLAN Routing & Server Reachability

So I think actually this was probably the most important test. From IT-PC1 (192.168.10.102 in VLAN 10) I pinged Admin-PC1 at 192.168.20.101 which is in VLAN 20. The first packet timed out and that's normal because the switch needs a moment to resolve MAC addresses through ARP. But the next three replies came back at less than 1ms which proves the Core.SW is routing between VLANs properly.

I also pinged the DNS Server at 192.168.40.20 from IT-PC1 and got successful replies, so the server room in VLAN 40 is reachable from other VLANs as well.

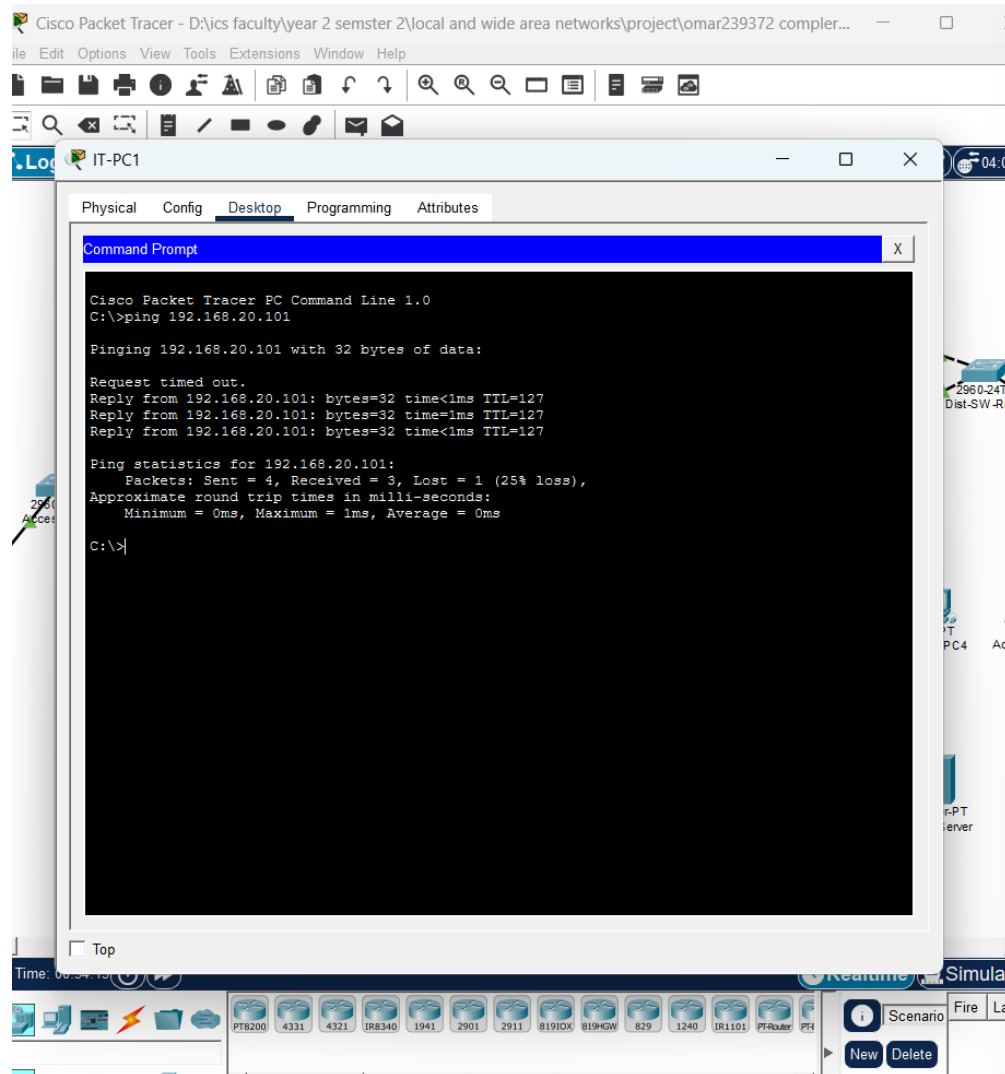


Figure 9 — IT-PC1 pinging Admin-PC1 at 192.168.20.101, inter-VLAN routing is working.

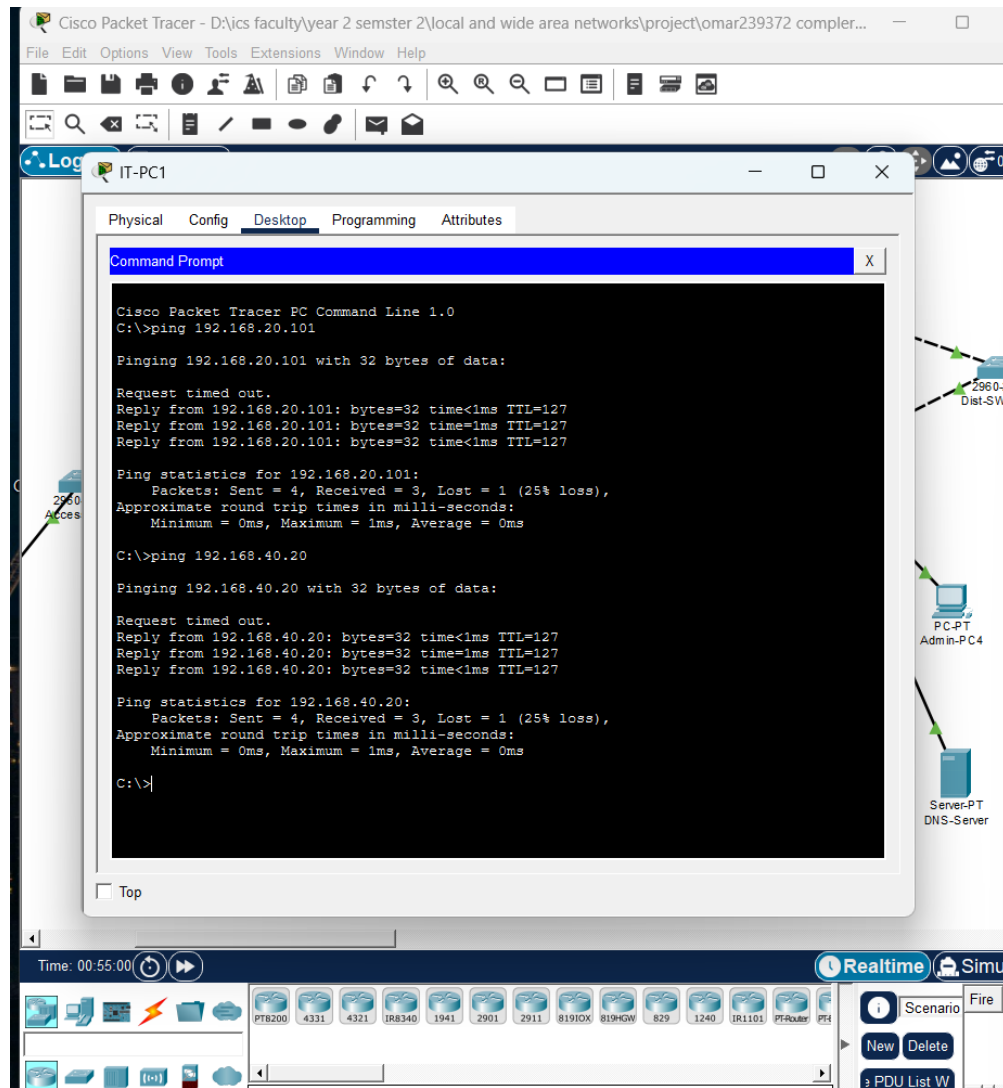


Figure 10 IT-PC1 pinging DNS-Server at 192.168.40.20, server room is reachable.

Web Browser Test

To test HTTP i opened the web browser on IT-PC1 and typed <http://192.168.40.30> in the URL. The Cisco Packet Tracer welcome page loaded succesfully with the message "Welcome to Cisco Packet Tracer" which proves the Web server is working fine and the path from VLAN 10 to VLAN 40 is fully functional.

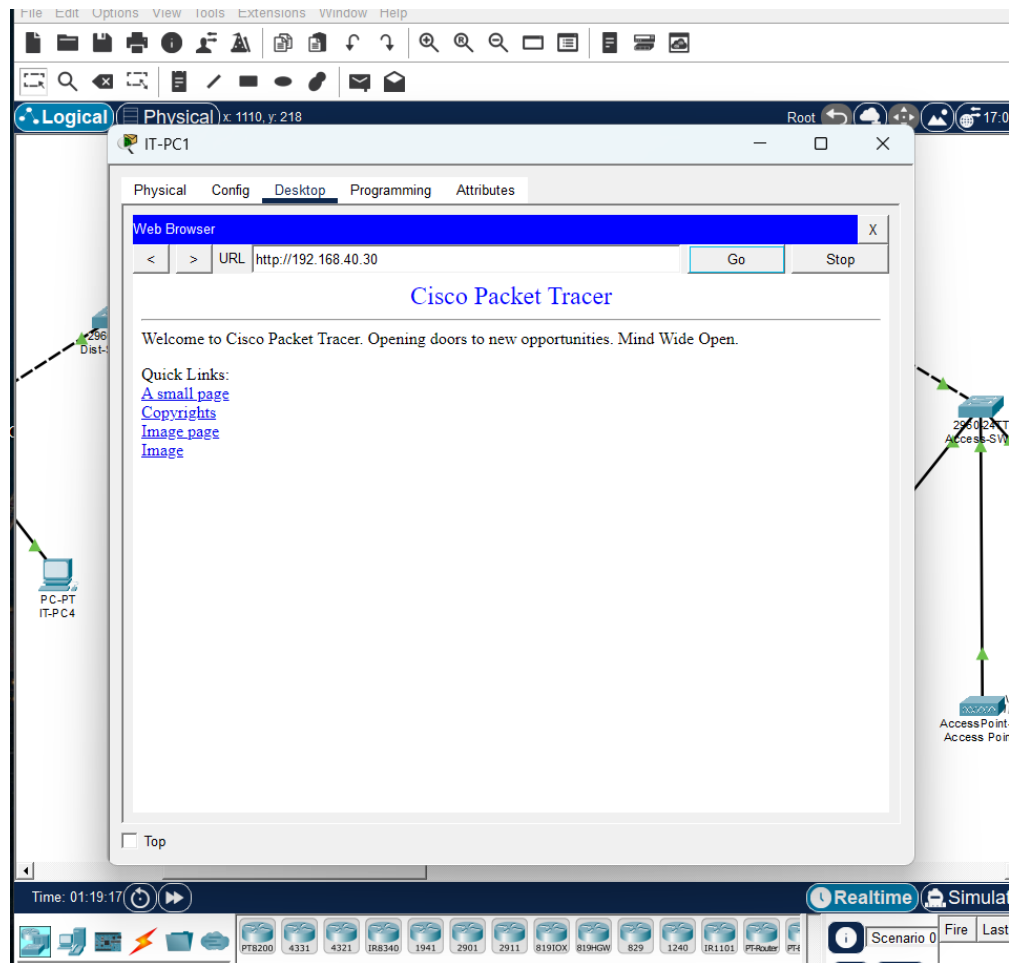


Figure 11 — IT-PC1 web browser loading <http://192.168.40.30> succesfully.

WiFi Connectivity

I connected the student Laptop1 connected to "University-WiFi" and got an IP through DHCP. I pinged IT-PC1 at 192.168.10.102 from the laptop and got 4 out of 4 replies with 0% loss. This shows that wireless devices in VLAN 30 can reach wired devices in VLAN 10 through the core switch.

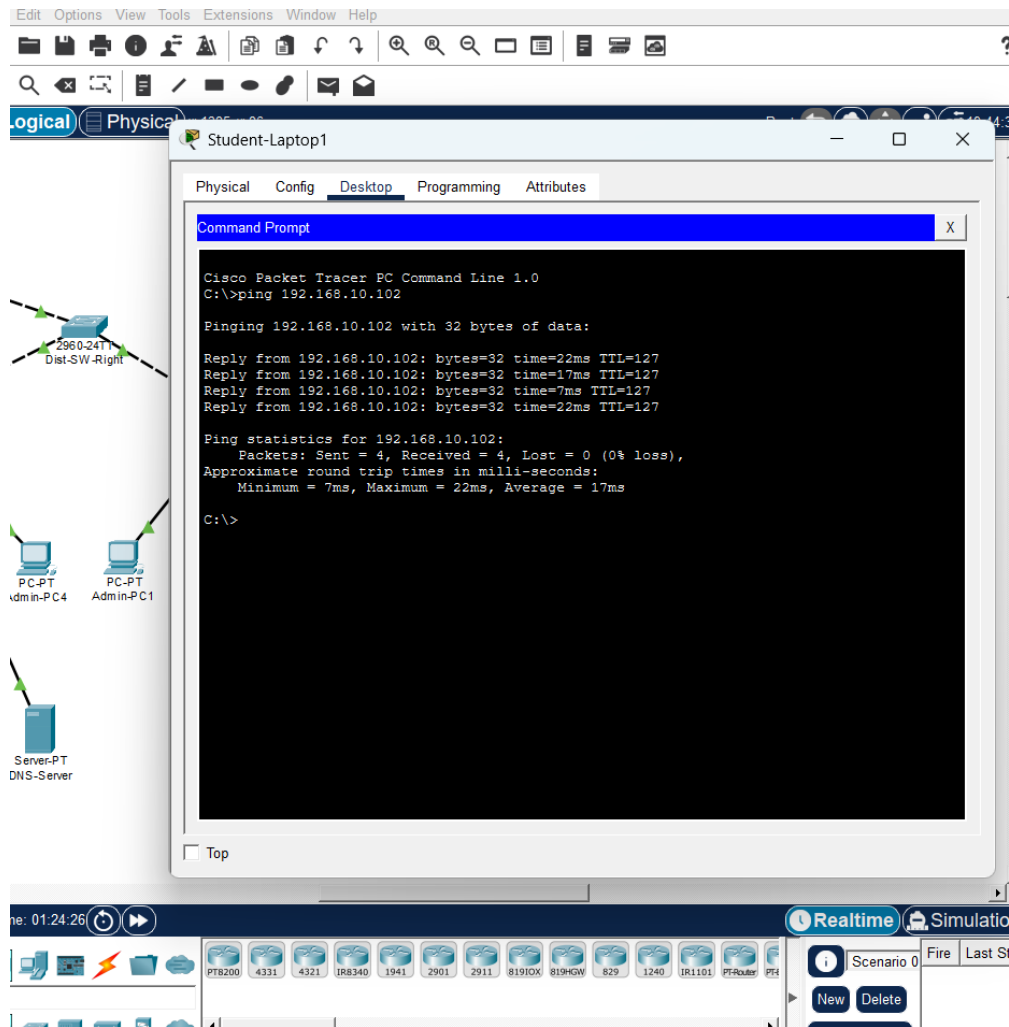


Figure 12 — Student-Laptop1 pinging IT-PC1 over WiFi, 4/4 replies with 0% loss.

Default Gateway Ping

Lastly i pinged the default gateway from IT-PC1 by typing ping 192.168.10.1. All four packets came back succesfully with 0% loss, confiring the SVI on Core-SW is reachable from the access layer which means basic connectivity is workig fine.

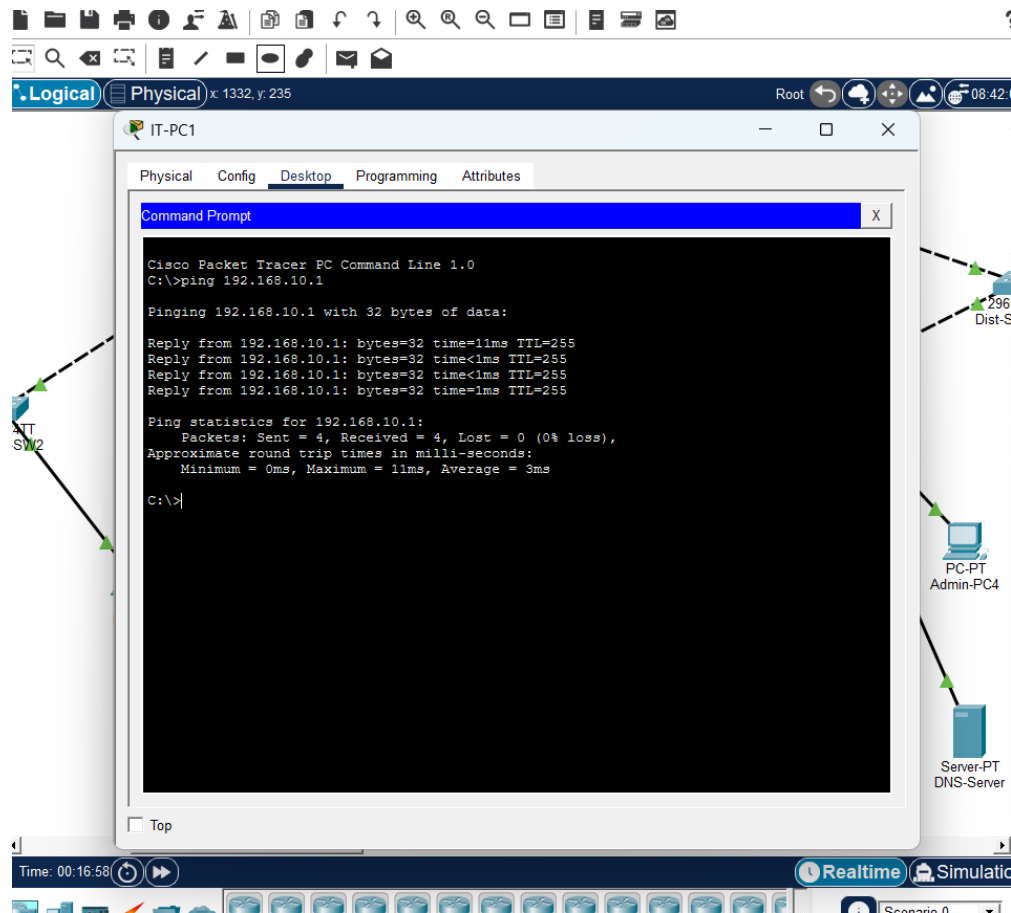


Figure 13 IT-PC1 pinging default gateway 192.168.10.1, 4/4 replies with 0% loss.

Conclusion & Reflection

So I think Overall this project went well. I built a working multi-VLAN LAN and all tests passed. The hardest part was getting inter-VLAN routing to work. I forgot to type "ip routing" on the Core-SW and spent ages troubleshooting before I realized. Such a small command but without it nothing gets routed. The DHCP part also confused me since the server is in VLAN 40 but clients are in VLANs 10 and 20, once the pools had correct gateways it worked immediately.

For improvements, I would add a redundant core switch so if Core-SW fails the network doesn't go down completely. Also ACLs on the core to restrict which VLANs can reach the servers, because right now any device can access everything which is a security risk. Forouzan [1] says segmentation without access control only solves half the problem.

This project gave me practical experience that just reading Kurose and Ross [4] couldn't give me. Typing commands and watching pings work or fail made concepts stick in my head way better than theory alone.

References

- [1] B A Forouzan Data Communications and Networking, 5th ed. New York, NY, USA McGraw-Hill, 2013.
- [2] added A STanenbaum and D J Wetherall Computr Netwrks 5th ed Upper Saddle River NJ USA Pearson, 2011.
- [3] IEEE Standards Association "IEEE Standard for Ethernet," IEEE Std 802.3-2018, 2018 Online Available: https://standards.ieee.org/standard/802_3-2018.html

4J F Kurose and K. W. Ross Computer Networking A TopDown Approach 7th ed. Boston MA USA Pearson, 2017.

5 ieee Standards Association "IEEE Standard for Information Technology Wireless LAN Medium Access Control and Physical Layer Specifications," IEEE Std 802.11-2020, 2020. Online]. Available: https://standards.ieee.org/standard/802_11-2020.html